CSA4701-DEEP LEARNING

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ASSIGNMENT 3 – GRAPHICAL MODEL EVALUATION

Assessment topic: Build a directed model (Bayesian network or HMM) and an undirected model (Markov random field) for a given domain. Specify factorization and independence assumptions, perform inference, and compare the models using log-likelihood, structural fit, and interpretability.

TASK 1 – Select and Define a Domain

Aim: To construct and compare a directed probabilistic graphical model and an undirected probabilistic graphical model for the same domain.

Illustrative domain: weather and whether a person carries an umbrella. Variables may include Weather (W), Rain (R) and Umbrella (U). Use a dataset or clearly stated probability values supplied by the instructor where available. The same variables and observations should be used for a fair comparison.

TASK 2 – Build the Directed Graphical Model

A Bayesian network represents variables as nodes in a directed acyclic graph (DAG). An arrow indicates a direct probabilistic dependency. For example, Weather → Rain → Umbrella, with Weather also influencing Umbrella if the domain supports that relationship.

The joint distribution factorizes as $P(W,R,U) = P(W)\,P(R|W)\,P(U|R,W)$. If the model assumes that Umbrella depends only on Rain, the factorization becomes $P(W,R,U) = P(W)\,P(R|W)\,P(U|R)$. This second form encodes the conditional independence $U \perp W \mid R$. State and justify whichever structure is selected.

TASK 3 – Build the Undirected Model

A Markov random field (MRF) uses an undirected graph. Its joint distribution is represented using non-negative potential functions over cliques: $P(X) = (1/Z) \prod_c \psi_c(X_c)$, where Z is the partition function that normalizes the distribution. For the illustrative domain, use pairwise potentials between Weather–Rain and Rain–Umbrella, or another structure justified by the data.

In an undirected graph, a node is conditionally independent of other nodes when the relevant separating nodes block all paths between them. State the separation assumption that follows from the chosen graph.

TASK 4 – Specify Conditional Independence Assumptions

For the Bayesian network Weather → Rain → Umbrella, the model states that Umbrella is conditionally independent of Weather once Rain is known, if no direct Weather → Umbrella edge is included. For the MRF, independence is read from graph separation rather than arrow direction. These are model assumptions and should be checked against the domain and data.

TASK 5 – Perform Inference

Inference means calculating a probability about one variable after observing other variables. For example, calculate P(Rain | Umbrella = yes). In a Bayesian network, use the chain rule and sum over hidden variables. For discrete variables: P(R|U) = P(R,U) / P(U), where P(R,U) is obtained by summing the joint distribution over all other variables.

For an MRF, compute the same conditional probability from the product of relevant clique potentials and normalize across possible values of Rain. Exact inference is feasible for a small graph; larger graphs may require approximate methods such as sampling.

TASK 6 – Evaluate the Models

Log-likelihood measures how well the model assigns probability to observed data: LL = Σ_i log P(x_i). A higher held-out log-likelihood (less negative) generally indicates better probabilistic fit. Use the same held-out observations for both models.

- Structural fit: compare whether the edges and independence assumptions are reasonable for the domain and whether the model captures observed relationships.
- Interpretability: explain how easily the graph and its conditional probabilities can be understood.
- Inference cost: record the method used and whether inference is exact or approximate.
- Avoid comparing log-likelihood values from differently defined data or incompatible evaluation settings.

TASK 7 – Record Results

Evaluation item	Directed model (BN/HMM)	Undirected model (MRF)
Graph structure	Describe chosen nodes and edges	Describe nodes and undirected edges

Factorization / potentials	Write factorization	Write clique-potential product
Conditional independence	State assumptions	State separation assumptions
Held-out log-likelihood	Enter measured value	Enter measured value
Inference example	Enter query and result	Enter query and result
Interpretability	Explain strengths/limits	Explain strengths/limits
Inference cost	Record method/time if measured	Record method/time if measured

TASK 8 – Compare and Defend the Choice

Compare both models using held-out log-likelihood, whether their structures match the domain, ease of explaining the assumptions, and inference cost. A directed model is useful when causal or directional relationships are natural to express. An undirected model is useful when relationships are mutual and no natural direction is needed. Do not claim one model is universally better; support the decision with the selected domain and measured results.

TASK 9 – Conclusion and References

Graphical models represent dependencies among random variables and support probabilistic inference. The final choice should balance fit, assumptions, computational cost and interpretability. Cite the dataset, framework and any reference used for the selected Bayesian network, HMM or MRF method.